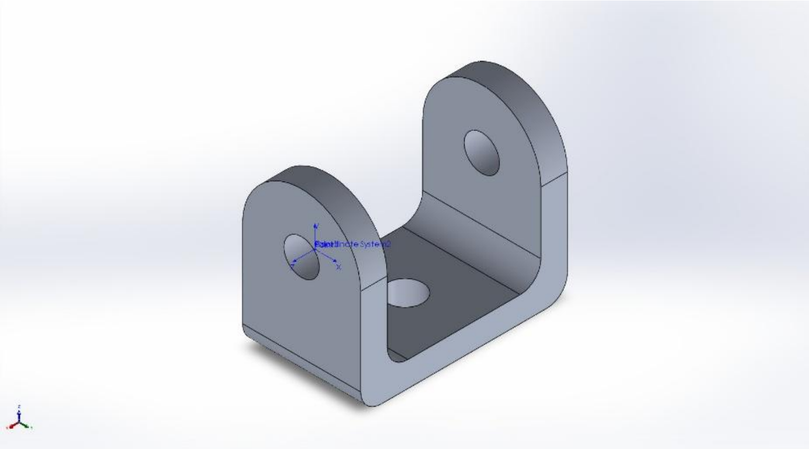


Simulation of U-Bracket for UAV Wing Attachment

Date: Sunday, November 16, 2025
Designer: Kevin Thomas Fernandez
Study name: Static Bracket Study
Analysis type: Static

Table of Contents

Description

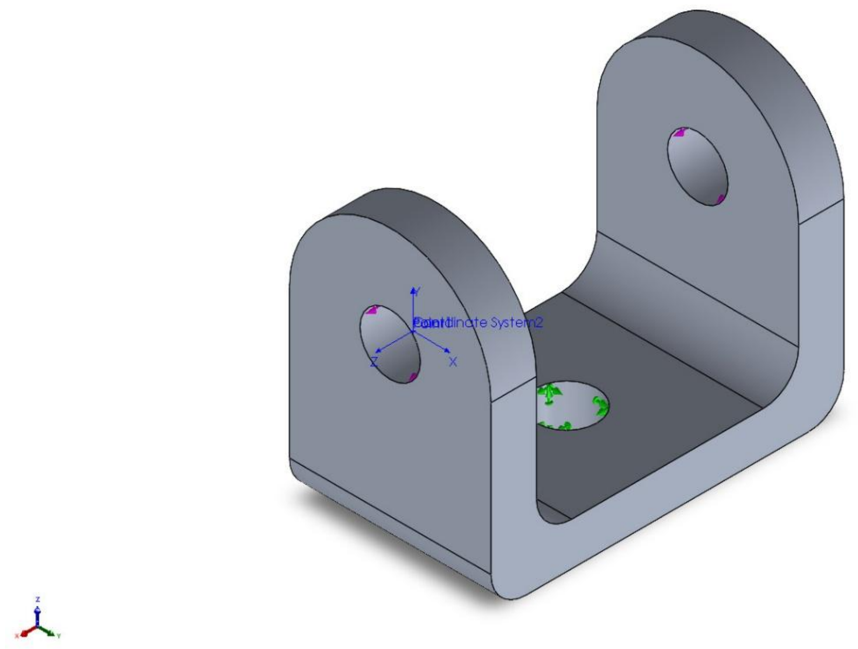
This study analyzes a U-shaped aluminum 7075-T6 bracket subject to a 500 N shear load. The objective is Evaluate stress and deflection of the U-bracket under a 500 N shear load.

Assumptions

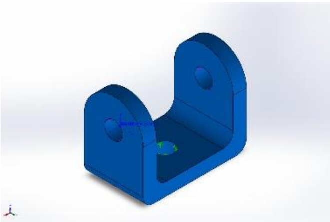
The static analysis assumes the bracket is made from linear elastic and isotropic 7075-T6 aluminum, with no plastic deformation, creep, or temperature-dependent behavior considered. The model represents a single solid body with all regions treated as perfectly bonded, and no frictional or contact interactions included. The base mounting face is idealized as fully fixed, preventing all translational and rotational motion, while a 500 N shear force is applied directly to the leg-hole surface to represent operational loading. Structural behavior is modeled with small strain assumptions, although large displacement mode is technically enabled; the load remains within the linear range. No bolt preload, thermal strain, aerodynamic pressure, or external moments are applied. The geometry is assumed to be perfectly machined, including ideal cylindrical holes and fillet transitions. A high-quality curvature-based tetrahedral mesh is used, and no distorted elements are present. This simulation represents a simplified, idealized loading case meant to evaluate bracket stiffness and maximum stress under the design load.

Kevin Thomas Fernandez
11/16/2025

Model Information



Model name: U-Bracket for UAV Wing Attachment
Current Configuration: Default

Solid Bodies			
Document Name and Reference	**Treated As**	**Volumetric Properties**	**Document Path/Date Modified**
Cut-Extrude4 	Solid Body	**Mass:0.146886 kg** **Volume:5.22727e-05 m^3** **Density:2,810 kg/m^3** **Weight:1.43948 N**	**U-Bracket for UAV Wing Attachment.SLDPRT** Nov 16 18:13:01 2025

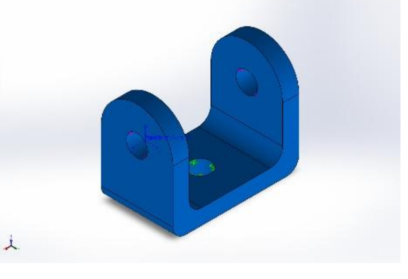
Study Properties

Study name	Static Bracket Study
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	On
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (C:\This Matters 2025\Engineering\Certificates\CSWA\SWEDU_CSWAExam_PracticeProblems_Levels)

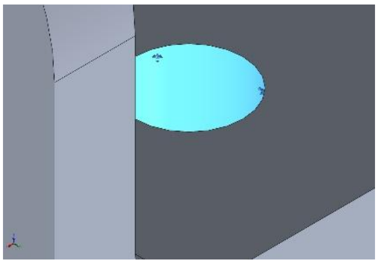
Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m^2

Material Properties

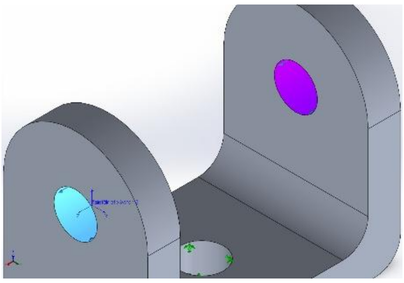
Model Reference	Properties	Components
	**Name:** 7075-T6 (SN) **Model type:** Linear Elastic Isotropic **Default failure criterion:** Unknown **Yield strength:** 5.05e+08 N/m^2 **Tensile strength:** 5.7e+08 N/m^2 **Elastic modulus:** 7.2e+10 N/m^2 **Poisson's ratio:** 0.33 **Mass density:** 2,810 kg/m^3 **Shear modulus:** 2.69e+10 N/m^2 **Thermal expansion coefficient:** 2.36e-05 /Kelvin	U-Bracket (U-Bracket for UAV Wing Attachment)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		**Entities:** 1 face(s) **Type:** Fixed Geometry

Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	-500.003	-0.00979519	0.00018692	500.003
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
Force-1		**Entities:** 1 face(s) **Reference:** Face< 1 > **Type:** Apply force **Values:** ---, ---, 500 N

Mesh information

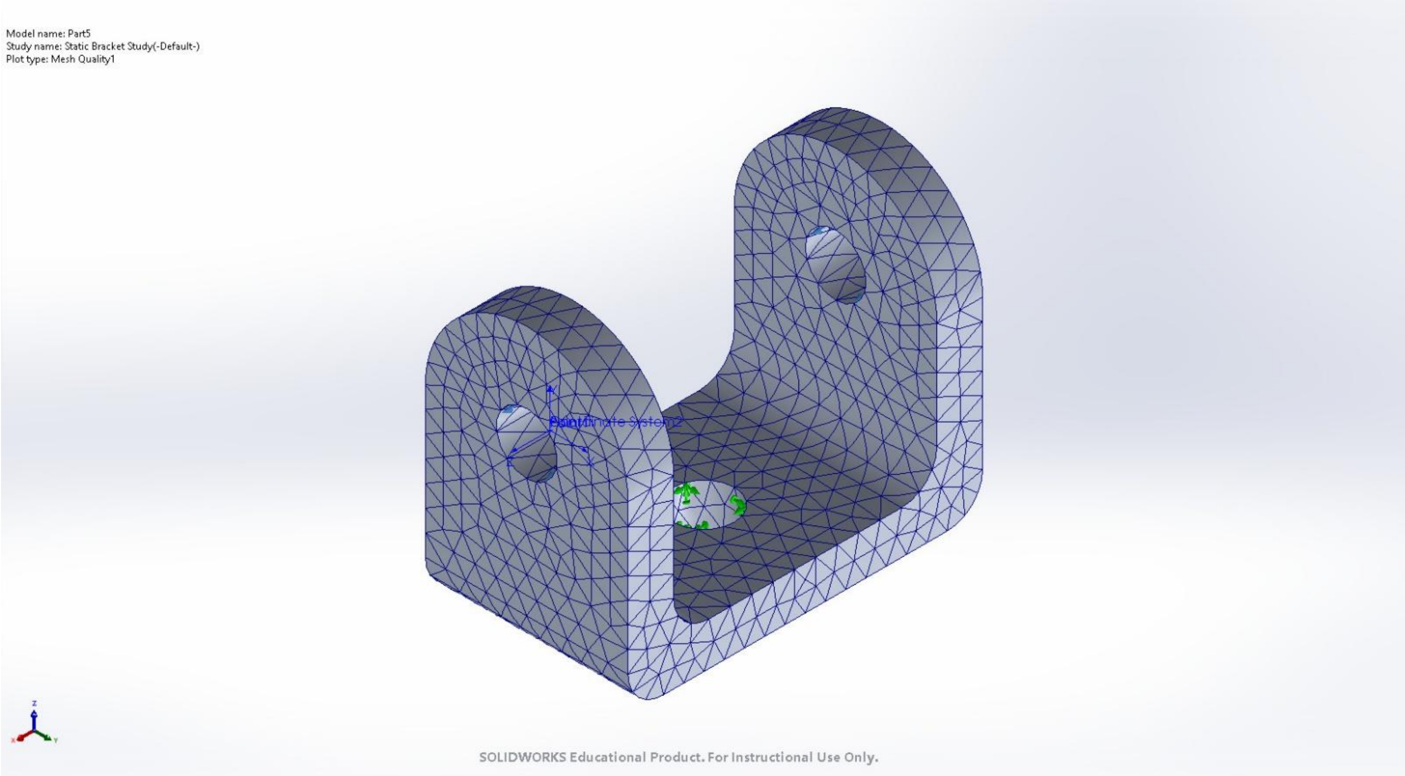
Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	4.07144 mm
Minimum element size	4.07144 mm
Mesh Quality	High

Mesh information - Details

Total Nodes	11693
Total Elements	7084
Maximum Aspect Ratio	4.4659
% of elements with Aspect Ratio < 3	99.4
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh;mm;ss):	00:00:01
Computer name:	STARS

Mesh Quality Plots

Name	Type	Min	Max
Quality1	Mesh	-	-



U-Bracket for UAV Wing Attachment-Static Bracket Study-Quality-Quality1

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-500.003	-0.00979519	0.00018692	500.003

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

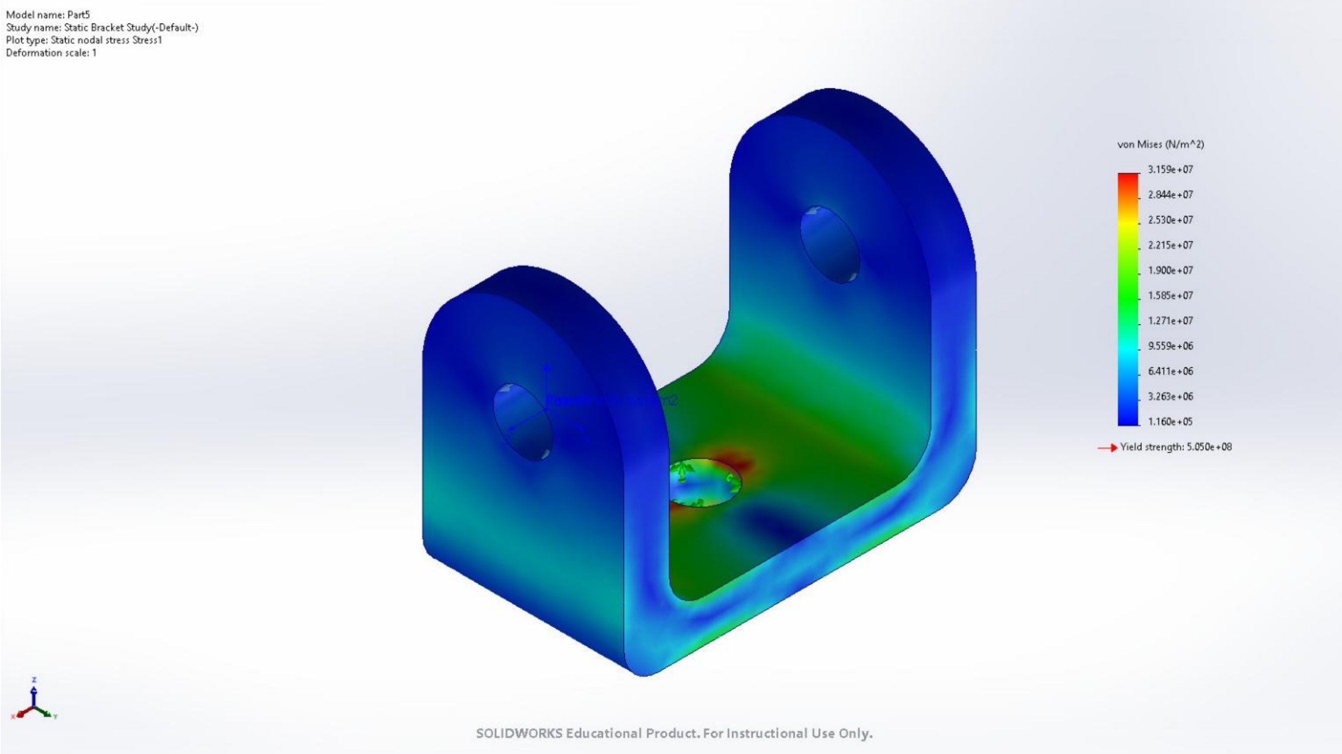
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	0	0	0	0

Free body moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

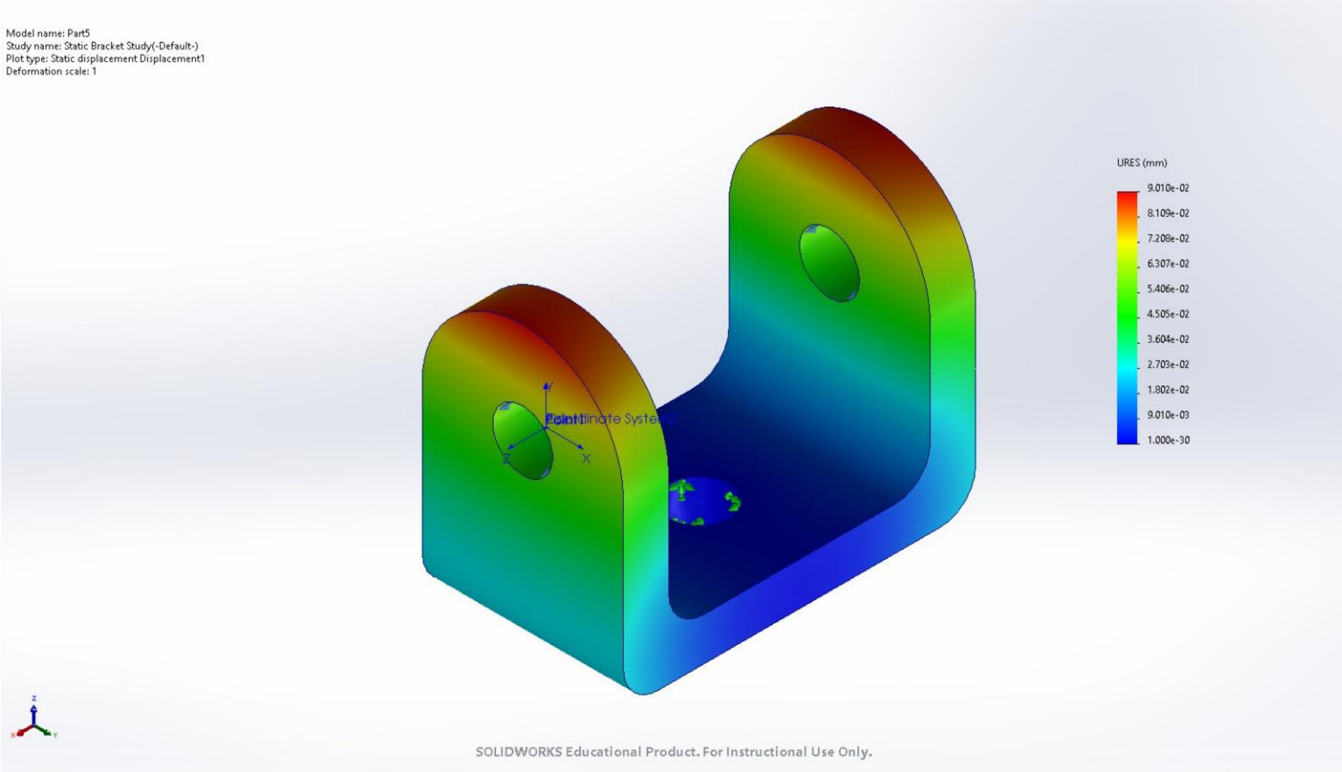
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	1.160e+05N/m^2 Node: 1049	3.159e+07N/m^2 Node: 5229



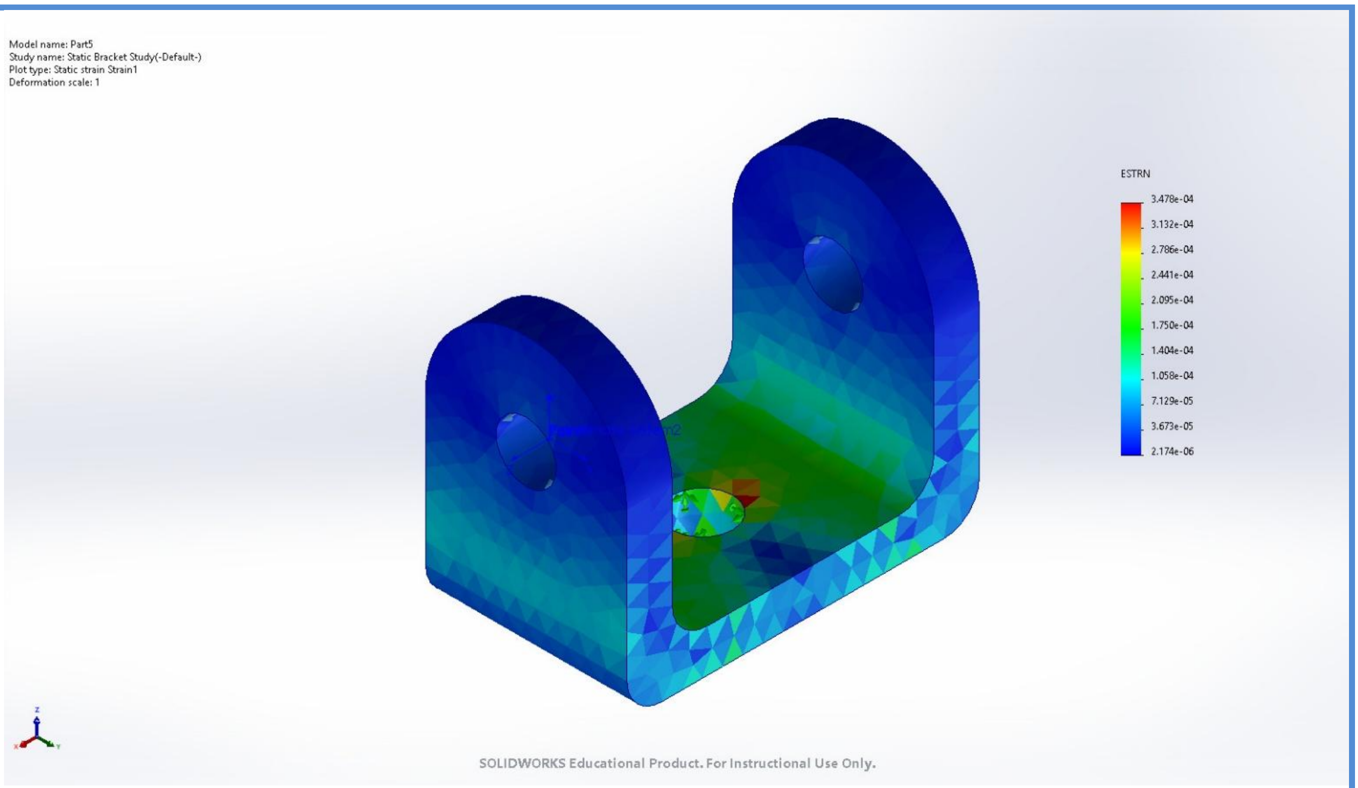
U-Bracket for UAV Wing Attachment-Static Bracket Study-Stress-Stress1

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 29	9.010e-02mm Node: 8769



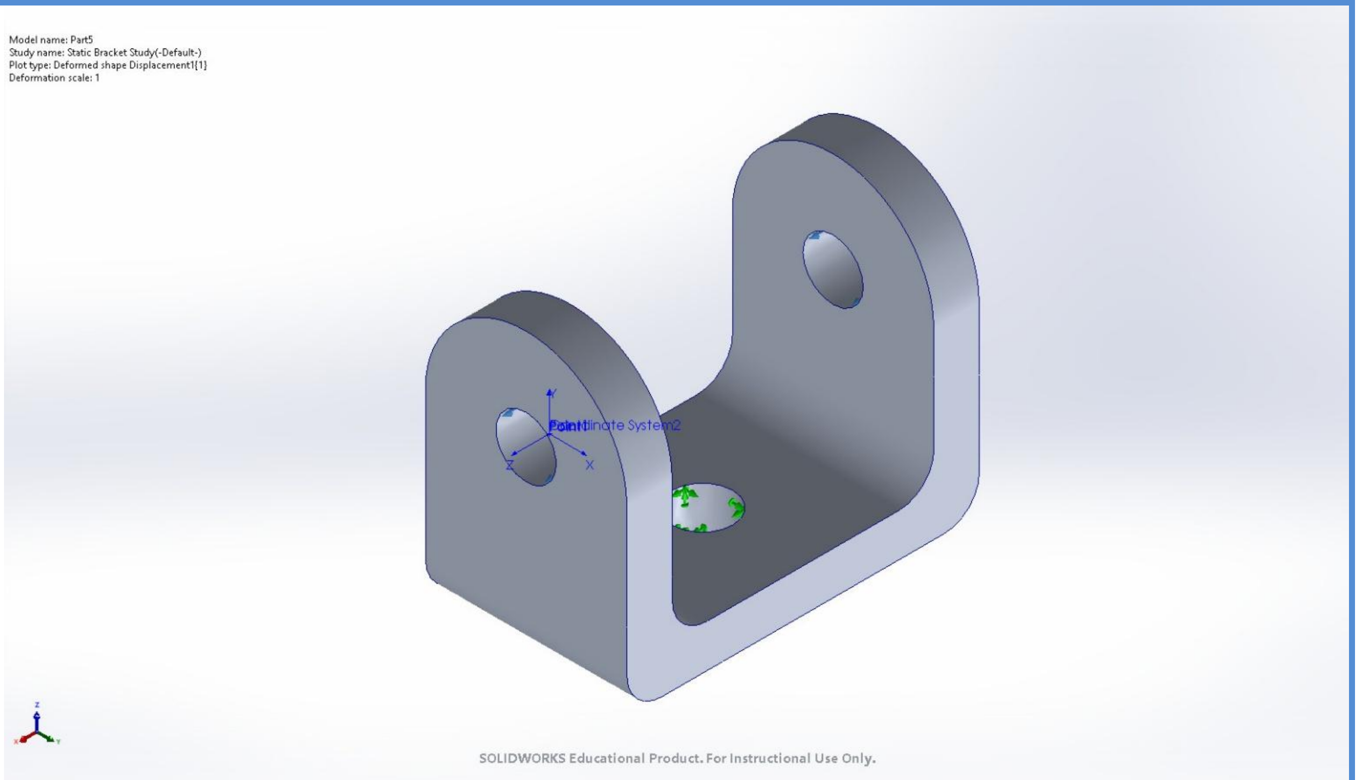
U-Bracket for UAV Wing Attachment-Static Bracket Study-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	2.174e-06 Element: 3689	3.478e-04 Element: 880



U-Bracket for UAV Wing Attachment-Static Bracket Study-Strain-Strain1

Name	Type
Displacement1{1}	Deformed shape



U-Bracket for UAV Wing Attachment-Static Bracket Study-Displacement-Displacement1{1}

Conclusion

The modal analysis shows that the bracket's first natural frequency is approximately 1684 Hz, which exceeds the minimum required 50 Hz by more than 33 times. This significant margin indicates that the bracket is exceptionally stiff and dynamically stable relative to expected UAV operational vibration environments. With all higher modes also occurring well above the 50 Hz threshold, the structure is not susceptible to resonance within any realistic flight conditions or propulsion-induced vibration spectra. Overall, the bracket meets and substantially surpasses all dynamic performance requirements and is fully suitable for integration into the UAV airframe.